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Display panel and display device

Abstract

A display panel and a display device are provided. The display panel includes a substrate and a color conversion layer. The substrate includes a first side surface and a second side surface disposed opposite to each other. A plurality of light-emitting areas are arranged in parallel on the first side surface, and a first positioning structure is disposed on the first side surface and is spaced apart from the light-emitting areas. A color conversion layer is disposed opposite to the first side surface of the substrate and includes a plurality of color conversion areas arranged in parallel and disposed corresponding to the light-emitting areas. A second positioning structure is disposed on a side of the color conversion layer facing the first side surface and arranged corresponding to the first positioning structure. The second positioning structure is connected to the first positioning structure.

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Background/Summary

RELATED APPLICATIONS

(1) This application claims the benefit of priority of Chinese Patent Application No. 202210921056.5, filed Aug. 2, 2022, the contents of which are incorporated by reference as if fully

set forth herein in their entirety.

BACKGROUND OF INVENTION

1. Field of Invention

(2) The present application relates to a technical field of displays, and particularly to a display panel and a display device.

2. Related Art

(3) With development of display technologies, quantum dot technologies are gradually favored by people. In display panels using quantum dot technologies, different color conversion areas on color conversion layers are coated with quantum dots of different colors, and then the color conversion layers are attached to backlight substrates, so that the quantum dots in the color conversion areas correspond to light-emitting areas on the backlight substrates, so as to achieve the display of the display panels. However, conventional color conversion layers and substrates are prone to displacement and misalignment during production or use, resulting in optical crosstalk between adjacent light-emitting areas, and adversely affecting display effects of display panels.

SUMMARY OF INVENTION

(4) Embodiments of the present application provide a display panel and a display device to overcome a problem that color conversion layers and substrates in conventional display panels are prone to displacement and misalignment, resulting in optical crosstalk.

(5) An embodiment of the present application provides a display panel, including a substrate comprising a first side surface and a second side surface disposed opposite to each other. A plurality of light-emitting areas are arranged in parallel on the first side surface, a first positioning structure is disposed on the first side surface and is spaced apart from the light-emitting areas, and a color conversion layer disposed opposite to the first side surface of the substrate and comprising a plurality of color conversion areas arranged in parallel and disposed corresponding to the light-emitting areas. A second positioning structure is disposed on a side of the color conversion layer facing the first side surface and arranged corresponding to the first positioning structure, and the second positioning structure is connected to the first positioning structure.

(6) Optionally, in some embodiments of the present application, the first positioning structure is protruded on the first side surface, the second positioning structure is protruded on the side of the color conversion layer facing the first side surface, and a side of the first positioning structure facing the second positioning structure is connected to a side of the second positioning structure facing the first positioning structure.

(7) Optionally, in some embodiments of the present application, a first positioning hole is located on the side of the first positioning structure facing the second positioning structure, and the second positioning structure is inserted into the first positioning hole, so that the second positioning structure is connected to the first positioning structure; and/or, a second positioning hole is located on the side of the second positioning structure facing the first positioning structure, and the first positioning structure is inserted into the second positioning hole, so that the first positioning structure is connected to the second positioning structure.

(8) Optionally, in some embodiments of the present application, the first side surface of the substrate is partially recessed for formation of the first positioning structure, and the second positioning structure is protruded on the side of the color conversion layer facing the first side surface and is inserted into the first positioning structure, so that the second positioning structure is connected to the first positioning structure; or the side of the color conversion layer facing the first side surface is recessed for formation of the second positioning structure, and the first positioning structure is protruded on the first side surface and is inserted into the second positioning structure, so that the first positioning structure is connected to the second positioning structure.

(9) Optionally, in some embodiments of the present application, the first positioning structure is located between adjacent ones of the light-emitting areas, and the second positioning structure is located between adjacent ones of the color conversion areas.

(10) Optionally, in some embodiments of the present application, there are first positioning structures and second positioning structures connected to the first positioning structures in a one-to-one correspondence.
(11) Optionally, in some embodiments of the present application, each of the first positioning structure and the second positioning structure is made of a thermoplastic material.
(12) Optionally, in some embodiments of the present application, each of the first positioning structure and the second positioning structure is made of a material comprising a magnetic material, and the side of the first positioning structure facing the second positioning structure and the side of the second positioning structure facing the first positioning structure magnetically attract each other.
(13) Optionally, in some embodiments of the present application, each of the first positioning structure and the second positioning structure is made of a material comprising a light-shielding material, there are first positioning structures arranged around a corresponding one of the light-emitting areas, and there are second positioning structures arranged around a corresponding one of the color conversion areas.
(14) Correspondingly, an embodiment of the present application further provides a display device including any one of the above-mentioned display panels, a casing connected to the display panel, and a control circuit disposed in the casing and electrically connected to the display panel.
(15) In the embodiment of the present application, the display panel includes the substrate and the color conversion layer. The substrate includes the first side surface and the second side surface disposed opposite to each other. The light-emitting areas are arranged in parallel on the first side surface, and the first positioning structure is disposed on the first side surface and is spaced apart from the light-emitting areas. The color conversion layer is disposed opposite to the first side surface of the substrate and includes the color conversion areas arranged in parallel and disposed corresponding to the light-emitting areas. The second positioning structure is disposed on the side of the color conversion layer facing the first side surface. The second positioning structure is disposed corresponding to the first positioning structure and is connected to the first positioning structure. By means of the corresponding connection between the first positioning structure and the second positioning structure, when the color conversion layer is attached to the substrate, the color conversion areas on the color conversion layer and the light-emitting areas on the substrate can be correspondingly arranged, thereby preventing optical crosstalk caused by displacement and misalignment of the color conversion areas and the light-emitting areas, thus improving the display effects of the display panel.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) In order to better illustrate the technical solutions in the embodiments of the present application, the following will briefly introduce the accompanying drawings that need to be used in the description of the embodiments. Apparently, the accompanying drawings in the following description show merely some embodiments of the present invention, and a person skilled in the art may still derive other drawings from these accompanying drawings without creative efforts.
(2) FIG. **1** is a schematic structural view of a display panel according to an embodiment of the present application.
(3) FIG. **2** is a schematic structural view of another display panel according to an embodiment of the present application.
(4) FIG. **3** is a schematic structural view of another display panel according to an embodiment of the present application.
(5) FIG. **4** is a schematic structural view of another display panel according to an embodiment of

the present application.
(6) FIG. **5** is a schematic structural view of another display panel according to an embodiment of the present application.
(7) FIG. **6** is a top plan view of a substrate according to an embodiment of the present application.
(8) FIG. **7** is a top plan view of another substrate according to an embodiment of the present application.
(9) FIG. **8** is a schematic structural view of a display device according to an embodiment of the present application.
DESCRIPTION OF PREFERRED EMBODIMENTS
(10) The technical solutions in the embodiments of the present application will be clearly and completely described below in conjunction with the drawings in the embodiments of the present application. Obviously, the described embodiments are only a part of the embodiments of the present application, but not all of the embodiments. Based on the embodiments in this application, all other embodiments obtained by those skilled in the art without creative work shall fall within the protection scope of this application. It should be understood that the specific embodiments described here are only used to illustrate the present application, and are not used to limit the present application. In this application, if no explanation is made to the contrary, the orientation words used, such as “upper” and “lower” usually refer to the upper and lower positions of the device in actual use or working state. Specifically, they refer to the direction of the drawings, and “inner” and “outer” refer to the outline of the device.
(11) Embodiments of the present application provide a display panel and a display device, which will be described in detail below. It should be noted that the description order of the following embodiments is not intended to limit the preferred order of the embodiments.
(12) Embodiments of the present application provides a display panel. As shown in FIGS. **1** to **5**, a display panel **100** includes a substrate **110** having a first side surface **111** and a second side surface **112** disposed opposite to each other. A plurality of light-emitting areas **113** are arranged in parallel on the first side surface **111** and are correspondingly provided with a light-emitting light source for the display panel **100**. In addition, a circuit structure is formed inside the substrate **110** and is correspondingly connected to the light-emitting light source in the light-emitting areas **113** to control the corresponding light-emitting light source to emit light, thereby regulating a display mode of the display panel **100**.
(13) Specifically, as shown in FIGS. **6** and **7**, a plurality of first positioning structures **114** are disposed on a first side surface **111** of the substrate **110** and are spaced apart from the light-emitting areas **113**. By forming the first positioning structures **114** on the first side surface **111** of the substrate **110**, it facilitates the positioning and connection between the substrate **110** and other functional film layers, thus preventing deviations in connection between other functional film layers and the substrate **110**. In addition, the disposition of the first positioning structures **114** spaced apart from the light-emitting areas **113** can prevent the first positioning structures **114** from negatively affecting light-emitting effects of the light-emitting areas **113**, which in turn adversely affects display effects of the display panel **100**.
(14) As shown in FIGS. **1** to **3**, the display panel **100** includes a color conversion layer **120** disposed opposite to the first side surface **111** of the substrate **110**. The color conversion layer **120** includes a plurality of color conversion areas **121** arranged in parallel and disposed corresponding to the light-emitting areas **113**. The color conversion layer **120** is mainly configured for color conversion of light emitted by the light-emitting light source in the light-emitting areas **113** on the substrate **110**, so that the light emitted by the light-emitting light source passes through the corresponding color conversion area **121** on the color conversion layer **120** and is converted into a target color to achieve different display requirements of the display panel **100**.
(15) Specifically, a plurality of second positioning structures **122** are disposed on a side of the color conversion layer **120** facing the first side surface **111** and arranged corresponding to the first

positioning structure **114**, and the second positioning structure **122** are connected to the first positioning structures **114**, respectively. By disposing the second positioning structures **122** on the color conversion layer **120**, the second positioning structures **122** can be correspondingly connected to the first positioning structures **114** when the color conversion layer **120** is attached to the substrate **110**, so that the color conversion areas **121** and the light-emitting areas **113** are in a one-to-one correspondence, thereby preventing optical crosstalk caused by displacement and misalignment of the color conversion areas **121** and the light-emitting areas **113**, which in turn affect the display effects of the display panel **100**.
(16) Since the color conversion areas **121** are arranged corresponding to the light-emitting areas **113**, the second positioning structures **122** are arranged corresponding to the first positioning structures **114**, and the first positioning structures **114** are spaced apart from the light-emitting areas **113**, that is, the second positioning structures **122** are also spaced apart from the color conversion areas **121**, thus preventing the second positioning structures **122** from negatively affecting light-emitting effects of the light-emitting areas **113**, which in turn adversely affects display effects of the display panel **100**.
(17) It should be noted that the light-emitting light source corresponding to the light-emitting areas **113** may be a blue light source, and the corresponding color conversion areas **121** on the color conversion layer **120** may include a red conversion area, a green conversion area, and a blue conversion area, so that the light emitted by a blue light source after passing through the color conversion layer **120** is red light, green light, and blue light, respectively.
(18) Specifically, a red conversion area corresponds to a coating of red quantum dots which emit red light when excited by the blue light source. A green conversion area corresponds to a coating of green quantum dots which emit green light when excited by the blue light source. A blue conversion area can be left uncoated, allowing the blue light source to pass directly through the area to form blue light.
(19) It should be noted that the blue conversion area can also be coated with blue quantum dots, and the light-emitting light source can also use other color light sources, which is not limited here. It only needs to ensure that the light-emitting light source of the light-emitting areas **113** can simultaneously form red light, green light, and blue light after passing through the color conversion areas **121** on the color conversion layer **120** to meet different display requirements of the display panel **100**.
(20) The display panel **100** in the embodiment of the present application includes the substrate **110** and the color conversion layer **120**. The substrate **110** includes opposite first side surface **111** and second side surface **11**. A plurality of light-emitting areas **113** are arranged in parallel on the first side surface **111** and a first positioning structure **114** is disposed on the first side surface **111**. The first positioning structure **114** is spaced apart from the light-emitting areas **113**. The color conversion layer **120** is disposed opposite to the first side surface **111** of the substrate **110** and includes a plurality of color conversion areas **121** arranged in parallel and disposed corresponding to the light-emitting areas **113**. A second positioning structure **122** is disposed on a side of the color conversion layer **120** facing the first side surface **111**. The second positioning structure **122** is disposed corresponding to the first positioning structure **114** and is connected to the first positioning structure **114**. By means of the corresponding connection between the first positioning structure **114** and the second positioning structure **122**, the color conversion areas **121** on the color conversion layer and the light-emitting areas **113** on the substrate **110** can be correspondingly arranged, thereby preventing optical crosstalk caused by displacement and misalignment of the color conversion areas **121** and the light-emitting areas **113**, which in turn affect the display effects of the display panel **100**.
(21) Optionally, as shown in FIGS. **1** to **3**, the first positioning structure **114** is protruded on the first side surface **111**, the second positioning structure **122** is protruded on the side of the color conversion layer **120** facing the first side surface **111**, and a side of the first positioning structure

114 facing the second positioning structure **122** is connected to a side of the second positioning structure **122** facing the first positioning structure **114**. That is, the first positioning structure **114** and the second positioning structure **122** are convex in shape. The first positioning structure **114** and the second positioning structure **122** not only can play a positioning role in aligning the light-emitting areas **113** with the color conversion areas **121** in a one-to-one correspondence, but also can play a supporting role for the substrate **110** and the color conversion layer **120**. By designing thicknesses of the first positioning structure **114** and the second positioning structure **122**, a size of a gap between the light-emitting area **113** and the corresponding color conversion area **121** can be adjusted, thus achieving adjustment of light-emitting angles of the display panel **100**.

(22) In some embodiments, as shown in FIG. **2**, a first positioning hole **1141** is located on the side of the first positioning structure **114** facing the second positioning structure **122**, and the second positioning structure **122** is inserted into the first positioning hole **1141**, so that the second positioning structure **122** is connected to the first positioning structure **114**. By forming the first positioning hole **1141** in the first positioning structure **114**, a position of the second positioning structure **122** relative to the first positioning structure **114** after the second positioning structure **122** is inserted into the first positioning hole **1141** is more stable, thereby preventing left-right displacement between the first positioning structure **114** and the second positioning structure **122** from being occurred due to their relative moving during production or use of the display panel **100**, so that a risk of crosstalk occurring after the light emitted by the light-emitting areas **113** passes through the color conversion areas **121** is reduced, thus improving the display effects of the display panel **100**.

(23) Specifically, an end of the second positioning structure **122** facing the first positioning structure **114** can be directly inserted into the first positioning hole **1141**, that is, an orthographic projection of the second positioning structure **122** on the first side surface **111** falls within an orthographic projection of the first positioning structure **114** on the first side surface **111**. Alternatively, the side of the second positioning structure **122** facing the first positioning structure **114** is protruded with a first positioning portion, and the first positioning portion is inserted into the first positioning hole **1141**, so that the orthographic projection of the second positioning structure **122** on the first side surface **111** coincides with the orthographic projection of the first positioning structure **114** on the first side surface **111**, which is conducive to the corresponding arrangement of the color conversion areas **121** and the light-emitting areas **113**, and the influence of the first positioning structure **114** and the second positioning structure **122** on the light-emitting effects of the light-emitting areas **113** is prevented.

(24) In other embodiments, as shown in FIG. **3**, a second positioning hole **1221** is located on the side of the second positioning structure **114** facing the first positioning structure **114**, and the first positioning structure **114** is inserted into the second positioning hole **1221**, so that the first positioning structure **114** is connected to the second positioning structure **122**. By forming the second positioning hole **1221** in the second positioning structure **122**, a position of the first positioning structure **114** relative to the second positioning structure **122** after the first positioning structure **114** is inserted into the second positioning hole **1221** is more stable, thereby preventing left-right displacement of the first positioning structure **114** and the second positioning structure **122** from being occurred due to their relative moving during production or use of the display panel **100**, so that a risk of crosstalk occurring after the light emitted by the light-emitting areas **113** passes through the color conversion areas **121** is reduced, thus improving the display effects of the display panel **100**.

(25) Specifically, an end of the first positioning structure **114** facing the second positioning structure **122** can be directly inserted into the second positioning hole **1221**, that is, the orthographic projection of the first positioning structure **114** on the first side surface **111** falls within the orthographic projection of the second positioning structure **122** on the first side surface **111**. Alternatively, the side of the first positioning structure **114** facing the second positioning

structure **122** is protruded with a second positioning portion, and the second positioning portion is inserted into the second positioning hole **1221**, so that the orthographic projection of the first positioning structure **114** on the first side surface **111** coincides with the orthographic projection of the second positioning structure **122** on the first side surface **111**, which is conducive to the corresponding arrangement of the color conversion areas **121** and the light-emitting areas **113**, and the influence of the first positioning structure **114** and the second positioning structure **122** on the light-emitting effect of the light-emitting area **113** is prevented.
(26) In still other embodiments, a first positioning hole **1141** is formed on the side of the first positioning structure **114** facing the second positioning structure **122**, and the side of the second positioning structure **122** facing the first positioning structure **114** is protruded with a first positioning portion. In this case, the side of the second positioning structure **122** facing the first positioning structure **114** is provided with a second positioning hole **1221**, and the side of the first positioning structure **114** facing the second positioning structure **122** is protruded with a second positioning portion. The first positioning portion is inserted into the first positioning hole **1141**, and the second positioning portion is inserted into the second positioning hole **1221**, so that the first positioning structure **114** and the second positioning structure **122** are connected.
(27) Such a structural design enables the first positioning structure **114** and the second positioning structure **122** to be engaged with each other, which helps to improve stability of the relative positioning of the first positioning structure **114** and the second positioning structure **122**, so that the misalignment of the light-emitting areas **113** and the color conversion areas **121** during the production or use of the display panel **100** is prevented, thereby reducing a risk of crosstalk occurring after the light emitted by the light-emitting areas **113** passes through the color conversion areas **121**, and improving the display effects of the display panel **100**.
(28) Optionally, as shown in FIG. **4**, the first side surface **111** of the substrate **110** is partially recessed to form the first positioning structure **114**, and the second positioning structure **122** is protruded on the side of the color conversion layer **120** facing the first side surface **111**. The second positioning structure **122** is inserted into the first positioning structure **114** to connect the second positioning structure **122** to the first positioning structure **114**. That is, the first positioning structure **114** is a positioning groove located on the first side surface **111** of the substrate **110**, and the second positioning structure **122** is a positioning portion protruding on the color conversion layer **120**. The positioning connection between the substrate **110** and the color conversion layer **120** is achieved by mutual cooperation between the positioning portion and the positioning groove, and at the same time, the misalignment of the light-emitting areas **113** and the color conversion areas **121** during the production or use of the display panel **100** can be prevented, thereby reducing the risk of crosstalk occurring after the light emitted by the light-emitting areas **113** passes through the color conversion areas **121**, and improving the display effects of the display panel **100**.
(29) In some embodiments, as shown in FIG. **5**, the side of the color conversion layer **120** facing the first side surface **111** is partially recessed to form the second positioning structure **122**, and the first positioning structure **114** is protruded on the first side surface **111**. The first positioning structure **114** is inserted into the second positioning structure **122** to connect the first positioning structure **114** to the second positioning structure **122**. That is, the second positioning structure **122** is a positioning groove located on the color conversion layer **120**, and the first positioning structure **114** is a positioning portion protruding on the first side surface **111** of the substrate **110**. The positioning connection between the substrate **110** and the color conversion layer **120** is achieved by the mutual cooperation between the positioning portion and the positioning groove, and at the same time, the misalignment of the light-emitting areas **113** and the color conversion areas **121** during the production or use of the display panel **100** can be prevented, thereby reducing the risk of crosstalk occurring after the light emitted by the light-emitting areas **113** passes through the color conversion areas **121**, and improving the display effects of the display panel **100**.
(30) In other embodiments, a positioning groove and a positioning portion are formed on the first

side surface **111** of the substrate **110** to form the first positioning structure **114**. The side of the color conversion layer **120** facing the first side surface **111** is also provided with a positioning groove and a positioning portion to form the second positioning structure **122**. In connection, the positioning portion in the first positioning structure **114** is inserted into the positioning groove in the second positioning structure **122**, and the positioning portion in the second positioning structure **122** is inserted into the positioning groove in the first positioning structure **114**, so that the first positioning structure **114** and the second positioning structure **122** can be engaged together, which helps to improve the stability of the relative positions of the first positioning structure **114** and the second positioning structure **122**, and the misalignment of the light-emitting areas **113** and the color conversion areas **121** during the production or use of the display panel **100** is prevented, thereby reducing the risk of crosstalk occurring after the light emitted by the light-emitting areas **113** passes through the color conversion areas **121**, and improving the display effects of the display panel **100**.
(31) Optionally, the first positioning structure **114** is located between adjacent two of the light-emitting areas **113**, and the second positioning structure **122** is located between adjacent two of the color conversion areas **121**. Since the first positioning structure **114** and the second positioning structure **122** are disposed correspondingly, the first positioning structure **114** is disposed between adjacent ones of the light-emitting areas **113**, and the second positioning structure **122** is arranged between adjacent ones of the color conversion areas **121**, it ensures that the relative arrangement between the light-emitting areas **113** and the color conversion areas **121** when the first positioning structure **114** is connected to the second positioning structure **122**, thereby reducing the risk of misalignment of the light-emitting areas **113** and the color conversion areas **121** during the production of the display panel **100**.
(32) In some embodiments, a first positioning structure **114** is disposed between any two adjacent light-emitting areas **113**, a second positioning structure **122** is disposed between any two adjacent color conversion areas **121**, and the first positioning structures **114** are connected to the second positioning structures **122** in a one-to-one correspondence. By means of the first positioning structure **114** arranged between any two adjacent light-emitting areas **113**, the second positioning structure **122** arranged between any two adjacent color conversion areas **121**, any one of the light-emitting areas **113** and a corresponding one of the color conversion areas **121** can be positioned in place through the corresponding first positioning structure **114** and the second positioning structure **122**, thereby further helping reduce the risk of misalignment of the light-emitting areas **113** and the color conversion areas **121** during the production of the display panel **100**.
(33) It should be noted that the setting positions and setting methods of the first positioning structure **114** and the second positioning structure **122** can be adjusted correspondingly according to actual production and use conditions, it only needs to ensure that the arrangement of the first positioning structure **114** and the second positioning structure **122** can prevent the misalignment of the light-emitting areas **113** and the color conversion areas **121** during the production or use of the display panel **100**, which is not specifically limited here.
(34) Optionally, materials used for the first positioning structure **114** and the second positioning structure **122** in the embodiment of the present application are thermoplastic. That is, during a manufacturing process of the display panel **100**, when the first positioning structure **114** is protruded on the first side surface **111** of the substrate **110** and the second positioning structure **122** is protruded on the side of the color conversion layer **120** facing the first side **111**, the first positioning structure **114** and the second positioning structure **122** can be melted by heating, and then integrated by cooling, so as to realize the connection between the first positioning structure **114** and the second positioning structure **122**.
(35) Specifically, when the side of the first positioning structure **114** facing the second positioning structure **122** is provided with the first positioning hole **1141** and the second positioning structure **122** is inserted into the first positioning hole **1141**, or when the side of the second positioning structure **122** facing the first positioning structure **114** is provided with the second positioning hole

1221 and the first positioning structure **114** is inserted into the second positioning hole **1221**, or when the side of the first positioning structure **114** facing the second positioning structure **122** is provided with the first positioning hole **1141**, the side of the second positioning structure **122** facing the first positioning structure **114** is protruded with the first positioning portion, the side of the second positioning structure **122** facing the first positioning structure **114** is provided with the second positioning hole **1221**, and the side of the first positioning structure **114** facing the second positioning structure **122** is protruded with the second positioning portion, and when the first positioning portion is inserted into the first positioning hole **1141** and the second positioning portion is inserted into the second positioning hole **1221**, since the first positioning structure **114** and the second positioning structure **122** are in engagement with each other, when the first positioning structure **114** and the second positioning structure **122** are heated, it is more favorable for integration of the first positioning structure **114** and the second positioning structure **122**, thereby improving the stability of the relative position of the first positioning structure **114** and the second positioning structure **122**.
(36) Optionally, materials used for the first positioning structure **114** and the second positioning structure **122** in the embodiment of the present application include magnetic materials. The side of the first positioning structure **114** facing the second positioning structure **122** and the side of the second positioning structure **122** facing the first positioning structure **114** magnetically attract each other. That is, the first positioning structure **114** and the second positioning structure **122** are made of magnetic materials. When the color conversion layer **120** is connected to the substrate **110**, the connection between the first positioning structure **114** and the second positioning structure **122** can be achieved by mutual attraction between the first positioning structure **114** and the second positioning structure **122**. Such an arrangement makes the connection between the first positioning structure **114** and the second positioning structure **122** more flexible and convenient, which is helpful for simplifying the manufacturing process of the display panel **100**.
(37) Specifically, the materials used for the first positioning structure **114** and the second positioning structure **122** may be magnetic materials, such as permanent magnets, and the specific material types can be selected according to design requirements, and no special limitation is made here. It is only necessary to ensure the effective attraction of the first positioning structure **114** and the second positioning structure **122**.
(38) Optionally, materials used for the first positioning structure **114** and the second positioning structure **122** in the embodiment of the present application include light-shielding materials. That is, when in a positioning process in a corresponding one of the light-emitting areas **113** and a corresponding one of the color conversion areas **121**, the first positioning structure **114** and the second positioning structure **122** can further block the light emitted by the light-emitting light sources in adjacent ones of the light-emitting areas **113** to further prevent optical crosstalk between the adjacent ones of the light-emitting areas **113**, thus improving the display effects of the display panel **100**.
(39) Specifically, there are positioning structures **114** arranged around the corresponding light-emitting areas **113**, and there are second positioning structures **122** arranged around the corresponding color conversion area **121**. That is, the first positioning structures **114** are located between and shield any two adjacent light-emitting areas **113**, and the second positioning structures **122** are located between and shield any adjacent two color conversion areas **121**. The first positioning structure **114** is connected to the second positioning structure **122** so as to shield the two adjacent light-emitting areas **113**, and prevent the crosstalk between light emitted by the light-emitting light sources in the two adjacent light-emitting areas **113**, thereby improving the display effects of the display panel **100**.
(40) It should be noted that in addition to the methods of connecting the first positioning structure **114** to the second positioning structure **122** through heating and melting and then cooling for fusion, or magnetic attraction in the embodiments of the present application, it is also possible to

apply an adhesive layer directly on each of the side of the first positioning structure **114** facing the second positioning structure **122** and the side of the second positioning structure **122** facing the first positioning structure **114**, so that the connection between the first positioning structure **114** and the second positioning structure **122** is achieved through bonding between the adhesive layers.
(41) In addition, the first positioning structure **114** and the second positioning structure **122** can also be directly connected by vacuum adsorption. Specifically, the specific connection method of the first positioning structure **114** and the second positioning structure **122** can be selected according to the actual design requirements, and no special limitation is made here. It only needs to ensure that the first positioning structure **114** and the second positioning structure **122** can be effectively connected to prevent the misalignment of the light-emitting areas **113** and the corresponding color conversion areas **121** during the production or use of the display panel **100**.
(42) In addition, an embodiment of the present application further provides a display device, including a display panel. A specific structure of the display panel refers to the above-mentioned embodiments. Since the display device adopts all the technical solutions of the above-mentioned embodiments, the display device has at least all the beneficial effects brought by the technical solutions of the above-mentioned embodiments, which will not be repeated here.
(43) As shown in FIG. **8**, the display device **10** includes a display panel **100**, a control circuit **200**, and a casing **300**. Specifically, the casing **300** is connected with the display panel **100** to support and fix the display panel **100**. The control circuit **200** is disposed in the casing **300** and is electrically connected to the display panel **100** to control the display panel **100** to display images.
(44) Specifically, the display panel **100** may be fixed to the casing **300** to form an integral body with the casing **300**. The display panel **100** and the casing **300** form a closed space for accommodating the control circuit **200**. The control circuit **200** can be the main board of the display device **10**. In addition, the control circuit **200** may also be integrated with one or more functional components, such as a battery, an antenna structure, a microphone, a speaker, a headphone interface, a universal serial bus interface, a camera, a distance sensor, an ambient light sensor, and a processor, etc., so that the display device **10** can be used in various application fields.
(45) It should be noted that the display device **10** is not limited to the above contents, and may also include other devices, such as cameras, antenna structures, fingerprint unlocking modules, etc., to expand its use range, which is not limited here.
(46) The display device **10** in the embodiment of the present application has a wide range of applications, including televisions, computers, mobile phones, and flexible displays and lighting, such as foldable and rollable displays, and wearable devices, such as smart bracelets, smart watches, etc., all fall within the scope of the application field of the display device **10** in the embodiments of the present application.
(47) The display panel and the display device provided in the embodiments of the present application are described in detail above. Specific examples are used in this article to illustrate the principles and implementation of the application, and the descriptions of the above examples are only used to help understand the methods and core ideas of the application; in addition, for those skilled in the art, according to the idea of the application, there will be changes in the specific implementation and the scope of application. In summary, the content of this specification should not be construed as a limitation of the application.

Claims

1. A display panel, comprising: a substrate comprising a first side surface and a second side surface disposed opposite to each other, wherein a plurality of light-emitting areas are arranged in parallel on the first side surface, and a first positioning structure is disposed on the first side surface and is spaced apart from the light-emitting areas; and a color conversion layer disposed opposite to the first side surface of the substrate and comprising a plurality of color conversion areas arranged in

parallel and disposed corresponding to the light-emitting areas, wherein a second positioning structure is disposed on a side of the color conversion layer facing the first side surface and arranged corresponding to the first positioning structure, and the second positioning structure is connected to the first positioning structure, wherein the first side surface of the substrate is partially recessed for formation of the first positioning structure, and the second positioning structure is protruded on the side of the color conversion layer facing the first side surface and is inserted into the first positioning structure, so that the second positioning structure is connected to the first positioning structure; or the side of the color conversion layer facing the first side surface is recessed for formation of the second positioning structure, and the first positioning structure is protruded on the first side surface and is inserted into the second positioning structure, so that the first positioning structure is connected to the second positioning structure.

2. The display panel of claim 1, wherein the first positioning structure is protruded on the first side surface, the second positioning structure is protruded on the side of the color conversion layer facing the first side surface, and a side of the first positioning structure facing the second positioning structure is connected to a side of the second positioning structure facing the first positioning structure.

3. The display panel of claim 2, wherein a first positioning hole is located on the side of the first positioning structure facing the second positioning structure, and the second positioning structure is inserted into the first positioning hole, so that the second positioning structure is connected to the first positioning structure; and/or, a second positioning hole is located on the side of the second positioning structure facing the first positioning structure, and the first positioning structure is inserted into the second positioning hole, so that the first positioning structure is connected to the second positioning structure.

4. The display panel of claim 2, wherein each of the first positioning structure and the second positioning structure is made of a thermoplastic material.

5. The display panel of claim 2, wherein each of the first positioning structure and the second positioning structure is made of a material comprising a magnetic material, and the side of the first positioning structure facing the second positioning structure and the side of the second positioning structure facing the first positioning structure magnetically attract each other.

6. The display panel of claim 2, wherein each of the first positioning structure and the second positioning structure is made of a material comprising a light-shielding material, a first positioning structure is disposed between any two adjacent ones of the light-emitting areas, and a second positioning structure is disposed between any two adjacent ones of the color conversion areas.

7. The display panel of claim 1, wherein the first positioning structure is located between adjacent ones of the light-emitting areas, and the second positioning structure is located between adjacent ones of the color conversion areas.

8. The display panel of claim 7, wherein a first positioning structure is disposed between any two adjacent ones of the light-emitting areas, and a second positioning structure is disposed between any two adjacent ones of the color conversion areas.

9. A display device, comprising: the display panel of claim 1; a casing connected to the display panel; and a control circuit disposed in the casing and electrically connected to the display panel.

10. The display panel of claim 1, wherein the first positioning structure is provided with a positioning groove and a positioning portion on the first side surface, and the second positioning structure is provided with a positioning groove and a positioning portion on the side of the color conversion layer facing the first side surface; and in response to the second positioning structure being connected to the first positioning structure, the positioning portion of the first positioning structure is inserted into the positioning groove of the second positioning structure, and the positioning portion of the second positioning structure is inserted into the positioning groove of the first positioning structure.

11. The display panel of claim 1, wherein the first positioning structure and the second positioning structure are directly connected with each other by vacuum adsorption.